Access VR

Senior Design

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Background

Abstract

Access VR is a product that the aims to streamline and aid physical therapy. It allows users to be able to practice various scenarios and movements in a safe and controlled virtual space, thereby improving their confidence and skills in real-world wheelchair navigation. This will hopefully expedite the recovery and learning process of using a wheelchair for newly wheelchair bound users. Additionally, this technology can be used to allow for wheelchair bounds users to have more accessibility in virtual environments as it replicates wheelchair movement more accurately than traditionally controllers.

Intended Audience

The intended audience for the VR wheelchair system includes individuals who use wheelchairs, whether due to temporary or permanent mobility impairments. This product is designed for both new wheelchair users who need to develop basic navigation skills and experienced users who wish to refine and expand their capabilities. Additionally, rehabilitation centers, physical therapy clinics, and organizations pro-viding support for wheelchair users can utilize this system for training and educational purposes. The system is adaptable to various user needs and skill levels, making it a versatile tool for a wide range applications in both individual and institutional contexts.

Key Requirements

Virtual Environment

The system includes a variety of carefully designed VR environments that simulate different real-life scenarios. These environments provide users with opportunities to practice and enhance their navigation skills in a safe and controlled setting. \item User Interface: The VR system features an intuitive user interface that allows users to easily interact with the virtual environment. The interface is designed to be accessible and user-friendly, ensuring that wheelchair users can navigate the VR world with ease.

Sensor Based modules

The system utilizes sensor-based modules that attach to the wheels of the wheelchair. These sensors capture the movement of the wheels and translate it into corresponding actions within the VR environment. This allows users to control their movement in the VR world by spinning the wheels in real life.

Customization and Adaptability

The system can be customized to meet the specific needs of individual users. This includes adjusting the sensitivity of the sensors, the difficulty of the VR environments, and other settings to ensure an optimal training experience.

Architectural Design

Headset Layer

The Headset layer is the layer that connects the user with the virtual environment. The player movement provides the user the ability to move around the environment. Hand tracking give the user the ability to control the UI and therefore interact with the virtual environment.

Data Processing Layer

The Data Processing layer is an intermediate layer connecting the wheelchair controller with the headset. The Arduino microcontroller receives all input data from the wheelchair and sends that data to the API. The API then translates the data for use by the headset systems.

Wheelchair Layer

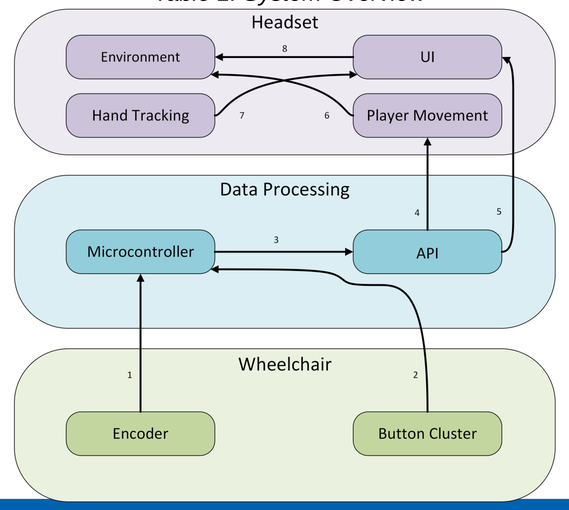
The Wheelchair layer is dedicated to the hardware aspect of this product. Which is all connected to the wheelchair so that information can be transferred to the data processing layer.

Key functions

User controller: Movement will be dependent on the movement of the wheels and buttons

Data transmission: Movement will be captured from the wheel and transferred to the micro-controller and or data processing layer.

Table 1. System Overview



Implementation Details and Test Plan

Hardware

- Wheelchair Model:
  - Feather Mobility - Red Cushion 13.5lb Silver Frame Lightweight Folding Wheelchair
- To capture Wheel movement:
  - LPD3806 Optical Rotary (Quadrature) Encoder
- Microprocessor
  - ELEGOO MEGA R3 Board ATmega 2560
- 3D printed Attachments made from provided senior design printers in ERB 335

Software

- VR Environment Development:
  - Unreal Engine by Epic Games
- VR Headset:
  - META Quest 3

Test Plan

- A user will sit in the wheelchair with the wheel propped up so that they will remain stationary while in use. They will be using the equipment to simulate real-life wheelchair movement into the VR Environment, and their goal is to navigate through different environments to simulate possible settings a wheelchair bound person might encounter.

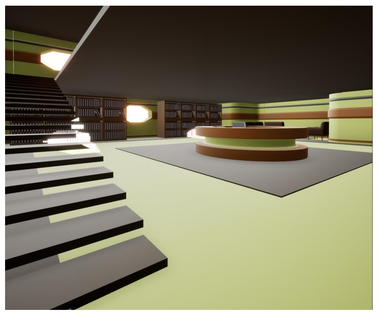


Figure 1. Level Design

Conclusions and Future Work

- We were able to create a working prototype with the help of the professor Dr. Gieser and the TA and lab coordinator Steve Mcdermott. This will be the starting ground for the next senior design team which will help the immersion of the project by potentially adding a tilt table and motors to simulate inclines.

References

1.Team Deliverables:

- https://github.com/Vidssk/AccessVR
- https://websites.uta.edu/cseseniordesign/

2.Related Works:

- McBride, Seth, et al. “Virtual Reality Project May Give New Wheelchair Users a Confidence Boost.” *New Mobility*, 1 Sept. 2021, www.newmobility.com/2017/09/virtual-reality-project-may-give-new-wheelchair-users-a-confidence-boost/.
- “Wheelsim - Wheelchair Simulator.” *Pretorian Technologies*, www.pretorianuk.com/wheelsim..

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